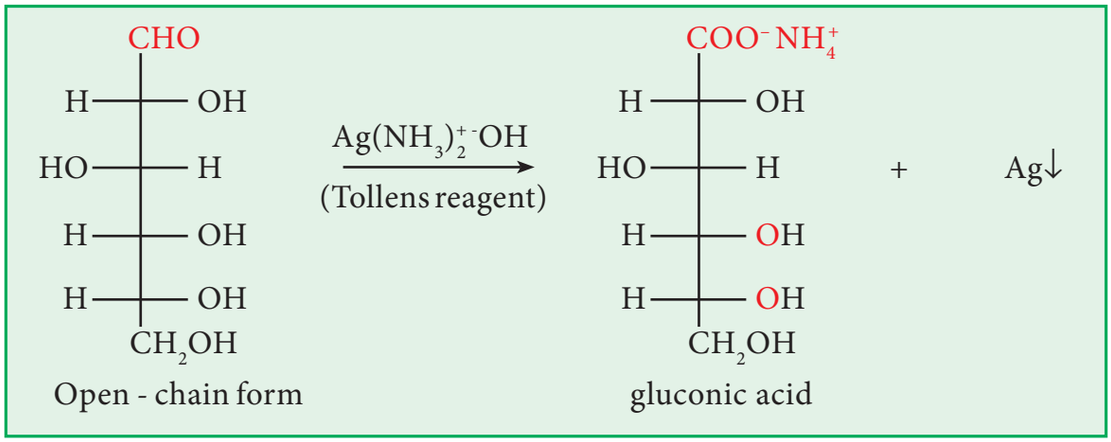
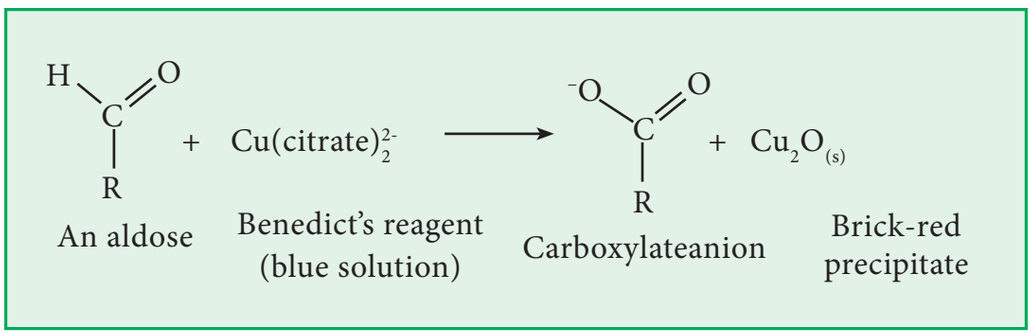


vii. Action with Fehling's solution:

Both glucose and fructose reduce Fehling's solution into red cuprous oxide.

viii. Action with Barfoed's and Benedict's reagents:

Both glucose and fructose reduce Barfoed's and Benedict's reagents into red cuprous oxide as in the case of Fehling's solution.



$$\text{Glucose} + CuO \rightarrow \text{Gluconic acid} + Cu_2O \downarrow$$

$$\text{Fructose} + CuO \rightarrow \text{Tartaric acid} + \text{glycollic acid} + Cu_2O \downarrow$$

Since both glucose and fructose reduce all these four reagents (Tollen's, Fehling's, Benedict's and Barfoed's reagent), these sugars are known as reducing sugars.

ix. Action with hydroxylamine:

a. Both glucose and fructose form oximes with hydroxylamine.

b. With NH_2OH, galactose forms galactoseoxime.

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